

Day 16: TQM Framework for Hyperlocal Logistics

Quality Engineering Solutions for **ZOMATO** and **SWIGGY**

1. Operational Challenges & Objective

Fast-paced delivery platforms like **ZOMATO** and **SWIGGY** depend heavily on speed and order correctness. During rush hours, hubs face food spillage from loose container lids, delivery delays, and wrong order handovers at pickup counters. Rather than fixing these issues separately, this study applies an integrated Total Quality Management framework to control process variation and reduce customer refunds.

2. Root Cause Diagnostic: Fishbone (6M) & Pareto 80/20 Rule

Using the 6M Fishbone framework, operational failures were mapped across six operational inputs:

- **Man (Riders and Kitchen Crew):** Riders rushing in traffic causing container drops. Kitchen staff handing over bags without verifying order IDs on the screen.
- **Material (Containers and Seals):** Non-vented lids opening under steam pressure and thin plastic containers cracking during transport.
- **Method (Dispatch SOPs):** Disorganized pickup tables causing bag mix-ups and high rider waiting times.
- **Machine / Tech:** App synchronization delay between merchant POS terminals and rider app screens.
- **Measurement:** Static SLA calculations failing to adapt during sudden order volume spikes.
- **Milieu (Environment):** Extreme weather conditions and traffic bottlenecks during lunch or dinner hours.

Pareto 80/20 Analysis: Defect tracking data shows that 80% of total order complaints come from 20% vital causes, specifically Man, Material, and Method failures. Directing corrective actions toward these three areas delivers the highest operational impact.

3. Statistical Process Control (p-Chart, np-Chart, c-Chart, Cp & Cpk)

To track order defects and process variation across delivery hubs, four statistical control tools are used based on sample sizes:

- **p-Chart (Fraction Defective):** Tracks the percentage of defective orders when daily batch sizes vary across different merchant hubs.
- **np-Chart (Number of Defectives):** Monitors the total count of defective orders when inspecting a fixed batch size (such as checking 500 orders daily per cluster).
- **c-Chart (Defects per Unit):** Tracks the total number of specific defects within a single order bag (e.g., item missing, gravy leakage, and missing cutlery in one order).
- **Process Capability (Cp & Cpk):** Evaluates if hub delivery times remain inside the target SLA window of 30 minutes (plus or minus 6 minutes). Maintaining Cpk above 1.33 ensures stable service delivery under peak load.

4. Visual Management & 3-Color Kanban Bins

Merchant pickup counters use visual cues and a 3-Color Kanban Bin system to standardize dispatch flow:

Kanban Bin Color	Food Category / Order Type	Visual Control & TQM Directive
RED BIN	High-Risk Liquids (Soups, Gravies, Ice Cream)	Mandatory double-sealing with anti-spill thermal wraps before QA inspection.
YELLOW BIN	In-Progress / Packing Check	Items packed and placed here for receipt item-count verification.
GREEN BIN	QA Verified / Ready for Dispatch	QR tamper seal applied. Rider picks directly from Green Bin with zero bag mix-ups.

5. Execution Plan: Poka-Yoke, 5S & Muda Elimination

- **Poka-Yoke (Mistake Proofing):** Smart QR tamper seals on delivery bags. The rider app blocks the "Order Picked Up" confirmation button until the rider scans the QR seal printed on the bag, preventing wrong handovers.
- **5S Workplace Setup:** Sorting and organizing merchant dispatch areas (Sort, Set in Order, Shine, Standardize, Sustain) to eliminate overcrowding at kitchen doors.
- **Muda (Waste) Elimination:** Reduces waiting waste for riders and overburden on kitchen crew, creating a smooth order dispatch flow without encouraging dangerous driving.

Summary: Implementing this structured TQM approach allows ZOMATO and SWIGGY to control process variation, prevent packaging defects, and maintain reliable delivery SLAs.